Lab Activity 2 – The Weather (Atmospheric Variables)

Purpose of Lab

The purpose of this lab activity is to familiarize students with the atmospheric variables measured by meteorologists worldwide.

In this lab activity, we will measure

- Relative Humidity
- Dew Point Temperature
- Ambient (Air) Temperature, Celsius and Fahrenheit
- Atmospheric (or Barometric) Pressure using four units

The Equipment

For this lab activity, we will use a self-contained “Weather Lab” module, which consists of an EPaper display (figure 1), an Arduino nano microcontroller (figure 2), a unified BME 280 sensor (figure 3), a 9V battery (figure 2), and a 3D printed custom “plastic” base (figure 1).

Turning the Weather Lab On

To turn the lab on, simply fully snap the power cap onto the 9V battery. The Arduino nano board will “light up”, and the EPaper display will blink off and on a few times.

Note that an EPaper display retains whatever information is written to it at the time the “Weather Lab” is powered off. This persistence of the EPaper display is a feature of the display, as well as an explanation of why the display is called an EPaper (or EInk) display. The display can be “left with last writing” permanently, without any damage to it.

Making Your Measurements

You will make measurements as a group. On your data sheet (at the end of this lab), write **your name**, and have your lab partners individually **write their names**. There should be only 3 or 4 individuals in a lab group.

Step 1: When you power on your ‘Weather Lab”, the display will show the following atmospheric measurement measurements for the classroom that you are in (figure 1):

- Rel Hum % (Relative Humidity in percent)
- Temp F deg (Room Temperature in Degrees Fahrenheit)
- Press mb (Atmospheric Pressure in millibars)
- Press psi (Atmospheric Pressure in pounds per square inch)

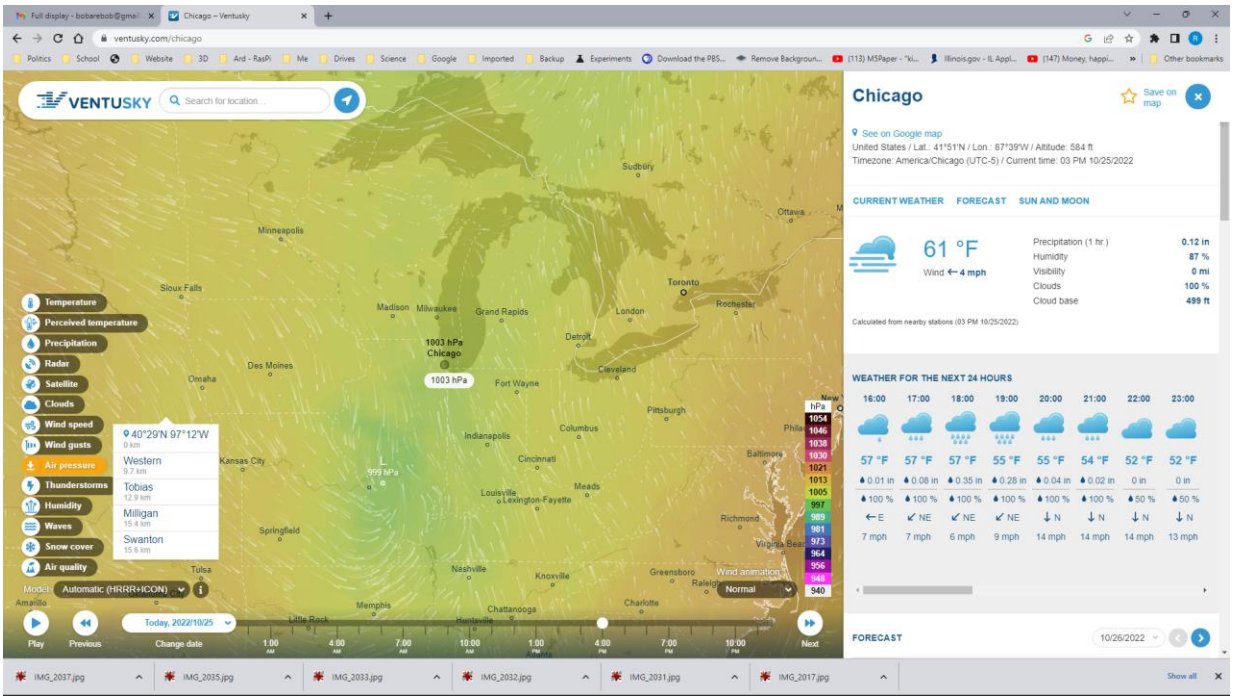
Write these readings into your data sheet in the appropriate place.

A millibar (mb) is one thousandth of a Barr, which is defined as 100,000 Pascals (Pa). Since atmospheric pressure at or near sea level is 101,325 Pa, it follows that the pressure unit One Atmosphere is 1.01325 Barr, or 1013.25 mb.

Now go the following website

https://www.ventusky.com/chicago

With a little effort, you should be able to see a map like this:



Be sure that the website is showing atmospheric pressure in **hectopascal** (hPa, which is the same as miilibar) for your location (which is Chicago). You can get the current reading for your location by hovering over your location on the website map.

(Hectopascal is the unit of Atmospheric Pressure used in Europe.)

Write down the atmospheric pressure measurement from the Ventusky website into the data sheet in the appropriate place.

Also **write down** the current outside Fahrenheit temperature, outside relative humidity, and altitude for Chicago, and transcribe these numbers into the data sheet in the appropriate place.

Finally, **answer these questions** in the data sheet:

- (Q1) Is the current atmospheric pressure less than or more than one standard atmosphere (1013.25 mb or hPa).
- (Q2) Is the sky cloudy (with possible rain), or is it relatively cloud free?

Step 2: On the left side of the EPaper display are four buttons called KEY1 through KEY4. We will press each button sequentially.

Press KEY1 and hold it pressed (figure 4). After a second or two, the EPaper display's first line will change to

- Dew Point F (Room Dew Point Temperature in Degrees Fahrenheit)

Write your reading into your data sheet in the appropriate place.

Now go to the following website:

https://www.calculator.net/dew-point-calculator.html?airtemperature=71.11&airtemperatureunit=fahrenheit&humidity=45.69&dewpoint=&dewpointunit=fahrenheit&x=47&y=18

Now type in the most current Fahreinheit room temperature and Relative Humidity to calculate the room's Dew Point Temperature.

Enter that calculated Dew Point Temperature value into you data sheet in the appropriate place, and **answer this question** in the data sheet:

- (Q3) Does the calculated Dew Point Temperature agree with the Weather Lab measured Dew Point Temperature?

Step 3: Press KEY2 and hold it pressed (figure 5). After a second or two, the EPaper display's second line will change to

- Temp C deg (Room temperature in Degrees Celsius)

Write your reading into your data sheet in the appropriate place.

Using the previously measured value of Fahrenheit room temperature, **answer this question** in the data sheet:

- (Q4) Do the measured Celsius (C) and Fahrenheit (F) room temperatures "fit" the following formula?

  C = 5/9 (F -32)

Step 4: Press KEY3 and hold it pressed (figure 6). After a second or two, the EPaper display's third line will change to

- Press inHg (Atmospheric Pressure in inches of Mercury)

Write your reading into your data sheet in the appropriate place.

In a standard barometer, the column height of liquid mercury (Hg) tells us quite accurately and precisely what the atmospheric pressure is. In the metric system, the column height of mercury corresponding to one atmosphere of air pressure is 76 cm or 760 mm.

Now **answer the following question** in the data sheet:

(Q5) Since there are 25.4 mm in every inch, how tall, in inches, is a 760 mm tall column of liquid mercury? The answer to this question tells us how many inches of mercury column height correspond to one standard atmospheric pressure of 1013.25 mb.

Step 5: Press KEY4 and hold it pressed (figure 7). After a second or two, the EPaper display's fourth line will change to

- Press Torr (Atmospheric Pressure in mm mercury or Torr)

Write your reading into your data sheet in the appropriate place.

A Torr is the unit equivalent of one mm of mercury column height, and so is $1/760^{th}$ of a standard atmospheric pressure of 1013.25 mb.

Now **answer the following question** in the data sheet:

- (Q6) Since one standard atmospheric pressure of 1013.25 mb, does this atmospheric measurement, in Torr, proportionately correspond to your previous measurement of atmospheric pressure in mb?

If time permits, your teacher will come by, with various devices, to manipulate your atmospheric environment.

Images

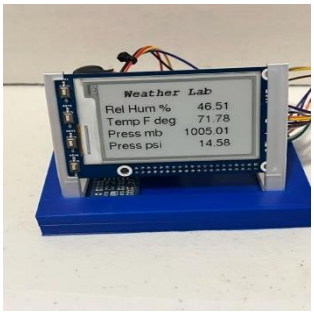


Figure 1

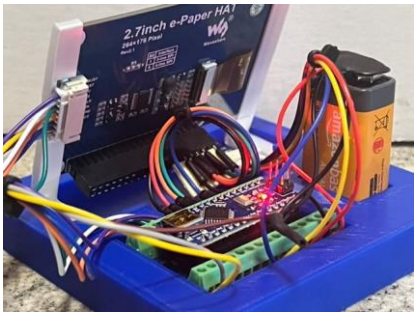


Figure 2

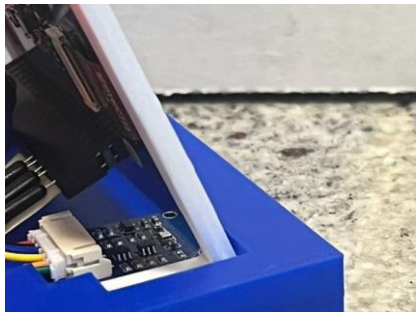


Figure 3

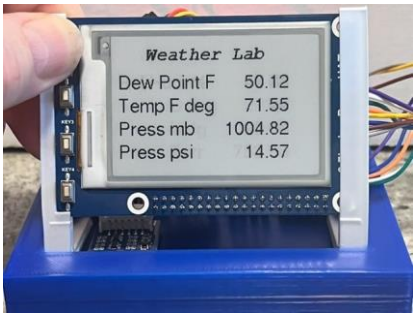


Figure 4

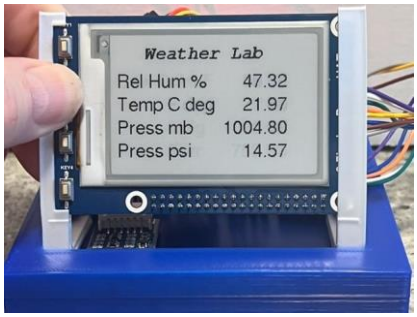


Figure 5

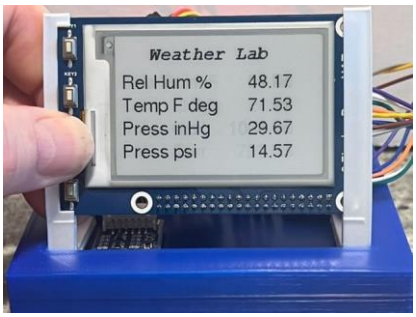


Figure 6

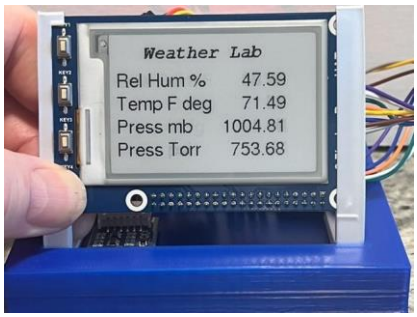


Figure 7

Group Data Sheet - Lab Activity 2 – The Weather (Atmospheric Variables)

Your Name: _____

Your Lab Partner's Names:

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Step 1 Readings:

Rel Hum % _____

Temp F deg _____

Press mb _____

Press psi _____

Ventusky website:

Outside atmospheric pressure, (hPa) _____

Outside temperature, Fahrenheit _____

Outside relative humidity, % _____

Altitude of Chicago above sea level, ft _____

Answer to Q1 (circle one) More Less

Answer to Q2 (circle one) Cloudy Cloud Free

*****

Step 2 Readings:

Measured Dew Point Temperature, F _____

Calculated Dew Point Temperature, F _____

Answer to Q3 (circle one) Yes No

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Step 3 Readings:

Room temperature, in degrees Celsius _____

Answer to Q4 (circle one) Yes No

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Step 4 Readings:

Atmospheric Pressure, inches Hg _____

Answer to Q5:

One standard atmosphere is _____ in of Hg

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Step 5 Readings:

Atmospheric pressure, in Torr _____

Answer to Q6: Yes No